

JACK NORTHROP

JACK NORTHROP (1895–1981) was born in Newark, New Jersey. Extraordinarily for a designer associated with pushing flight technology to its limits, he had no training as an engineer beyond high-school physics. He drifted into aircraft design in 1916, working for the Loughhead brothers in their workshop in Santa Barbara, California. After a spell with Douglas, Northrop helped Allan Loughhead found the Lockheed Aircraft Company in 1926. The following year, Northrop designed the Lockheed Vega, an airplane that exemplified his taste for radical and elegant design solutions. In 1929 he left to found the Northrop Aircraft Company, where he produced the Alpha, an airplane regarded as ahead of its time. Over the next 25 years he developed his pet “flying-wing” project. Although his flying-wing bombers such as the XB-35 were not adopted by the USAF – a personal catastrophe – their visionary design was later incorporated into stealth-aircraft designs.

DESIGN MEETING

Jack Northrop (right) is featured discussing designs with the Northrop Aircraft Company's assistant chief of design Walt Cerny (left) and project engineer Tom Quayle.



improved in particular by the adoption of ethylene glycol as a coolant – with a low freezing and high boiling point, it allowed radiators to be made smaller, reducing weight and drag. By the late 1930s, aircraft engines were capable of delivering well in excess of 1,000hp. Multiengine aircraft were now seriously powerful machines.

Modern refinements

Experiments at the NACA wind tunnel revealed that a fixed undercarriage contributed an astonishing 40 percent of the entire drag acting upon an airplane. Some designers responded by enclosing fixed undercarriages in streamlined “trousers,” but retractable undercarriages inevitably became standard in the course of the 1930s. Initially, there were considerable safety concerns about the possibility of the undercarriage failing to extend for landing. This is why in the Douglas DC-3, the most successful aircraft of the decade, the retracted landing gear still protruded below the fuselage so that, if the undercarriage failed to extend, the airplane could still come down on its wheels.

Other improvements included variable-pitch and then constant-speed propellers. Before this innovation, the setting of the propeller had to be

optimized for one stage of the airplane's flight at the expense of others. For example, one of Lindbergh's problems taking off for his transatlantic journey in the heavily laden Spirit of St. Louis was that his propeller was set to be optimal for cruising, not for takeoff. Another innovation was the use of flaps to temporarily change the shape of the aircraft's wing so that, for example, an aircraft designed for high speed would have enough lift to fly at lower speed when coming in to land. Safety was improved by attaching deicers to leading wing edges – initially inflatable rubber devices that punched the ice so that it cracked.

WIND-TUNNEL TESTING

This image taken in the test chamber at the Langley Research Center, Virginia, in 1932, shows the massive scale of its 30 x 60-ft (9 x 18-m) wind tunnel. A 31ft- (9.4m-) span Loening XSL-1 seaplane can be seen mounted on the test rack.

